

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 4: Decompression Sickness & Barotrauma

Syllabus Source: CAE Oxford (pp. 83-108) | Pass Target: >= 90% AviationExam

1. DECOMPRESSION SICKNESS (DCS) & HENRY'S LAW

Under normal atmospheric pressure, nitrogen gas is held in physical solution in body fluids and fatty tissues. When ambient barometric pressure decreases rapidly (above FL 180, and especially above FL 300), the body tissues become supersaturated, and nitrogen comes out of solution forming gaseous bubbles.

2. THE FOUR CLASSICAL CLINICAL FORMS OF DCS

DCS Clinical Form	Incidence & Severity	Typical Symptoms & Clinical Manifestations
The Bends (Joint & Muscular)	85% of cases	Deep, dull, throbbing pain in joints (knees, shoulders, elbows). Pain increases on movement. Relieved temporarily by firm pressure.
The Chokes (Pulmonary)	Rare (< 2%), but FATAL	Bubbles lodge in pulmonary capillary bed. Symptoms: Substernal burning chest pain, severe paroxysmal dry cough, dyspnoea, cyanosis, circulatory collapse.
The Creeps (Cutaneous / Skin)	10% of cases	Nitrogen bubbles in dermal capillaries. Symptoms: Intense itching, burning, tingling ('ants crawling under skin' - formication), mottled reddish/purplish skin rash.
The Stagers (Neurological)	Severe / Critical	Bubbles in central nervous system or spinal cord. Symptoms: Visual scotoma, severe headache, confusion, ataxia, hemiplegia, stroke-like focal deficits.

[*] OFFICIAL DEFINITION - Emergency Treatment of In-Flight Decompression Sickness

1. Immediate emergency descent to sea level or the lowest safe altitude; 2. Administer 100% OXYGEN via tightly fitting mask (accelerates nitrogen washout); 3. Immobilize affected limbs (do NOT exercise the affected joint); 4. Urgent transfer to a medical facility equipped with a HYPERBARIC RECOMPRESSION CHAMBER.

3. SCUBA DIVING RULES BEFORE FLYING (EASA PART-MED / ICAO)

[!] EXAM TRAP (AVIATIONEXAM) - Mandatory Wait Times Between Diving and Flying (Memorize 100%)

Scuba diving loads body tissues with dissolved nitrogen under high hyperbaric pressure. Flying afterward greatly multiplies DCS risk! - Non-Decompression Diving (Depth < 10 m / 30 ft, no decompression stops): Wait at least 12 HOURS before flying. - Decompression Diving (Requiring decompression stops) OR Multiple Dives over consecutive days: Wait at least 24 HOURS before flying. - Deep Technical / Saturation Diving: Recommended wait time is 48 HOURS.

4. BAROTRAUMA (BOYLE'S LAW $P \times V = \text{CONSTANT}$)

Barotrauma is physical tissue damage caused by expansion or contraction of trapped gas inside rigid body cavities when pressure changes cannot equalize:

Type of Barotrauma	Anatomical Site	Mechanism, Peak Danger Phase & Symptoms
Otic Barotrauma (Ear Block)	Middle ear via Eustachian Tube	Ascent: Air vents passively out easily. Descent: Eustachian tube acts as flap valve; if inflamed (cold/rhinitis), tube locks shut. Result: Severe pain, eardrum retraction, hematoma, rupture. Prevention: Valsalva, swallowing.
Sinus Barotrauma (Barosinusitis)	Frontal & Maxillary Sinuses	Blocked ostia trap air. Extreme sharp pain over forehead/cheeks on descent. May cause nosebleeds (epistaxis).
Dental Barotrauma (Barodontalgia)	Tooth pulp / defective cavity	Air pocket trapped under filling or abscess. Severe sharp toothache, typically occurs on ASCENT.
Gastrointestinal Barotrauma	Stomach, Small & Large Intestine	Gas expands on ascent (doubles by FL 180). Causes severe abdominal distension and cramping. Prevent by avoiding carbonated drinks and gas-producing food.

[!] EXAM TRAP (AVIATIONEXAM) - Ascent vs Descent: When is Barotrauma Worst?

- Ear Barotrauma (Otic) and Sinus Barotrauma occur almost exclusively during DESCENT, because air cannot enter swollen passages from the higher outside pressure. - Dental Barotrauma (Barodontalgia) and Gastrointestinal Distension occur predominantly during ASCENT, as trapped air expands.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Following a scuba dive requiring decompression stops, what is the minimum required waiting time before

- [A]: 4 hours
- [B]: 12 hours
- [C]: 24 hours
- [D]: 48 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. EASA Part-MED mandates at least 24 hours wait time after decompression diving (or multiple dives). For single non-decompression dives < 10 m, the minimum is 12 hours.

Q2: Which manifestation of Decompression Sickness is characterized by substernal burning chest pain, dry ha

- [A]: The Bends
- [B]: The Creeps
- [C]: The Chokes

- [D]: The Staggers

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. The Chokes is pulmonary DCS caused by gas bubbles occluding the pulmonary microcirculation; it is a life-threatening medical emergency.

Q3: During which phase of flight is a pilot most susceptible to acute otic barotrauma (ear block)?

- [A]: Rapid climb after take-off
- [B]: Steady high-altitude cruise
- [C]: Rapid descent from cruising altitude
- [D]: Engine start on the ground

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. During descent, ambient pressure rises and the Eustachian tube may collapse shut, preventing pressure equalization in the middle ear.

Q4: What gas law governs the formation of nitrogen bubbles in blood and joint tissues in Decompression Sickness?

- [A]: Boyle's Law
- [B]: Henry's Law
- [C]: Dalton's Law
- [D]: Charles' Law

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Henry's law governs the solubility of gases in liquids under varying ambient pressures.

Q5: What is the most common clinical manifestation of Decompression Sickness in aviation personnel?

- [A]: The Chokes
- [B]: The Bends (joint and deep muscle pain)
- [C]: Permanent paralysis
- [D]: Cardiac arrest

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The Bends represents ~85% of all DCS cases, causing dull throbbing pain in large joints (knees, elbows, shoulders).

Q6: A pilot who is suffering from a head cold flies a sector and experiences excruciating frontal headache during descent.

- [A]: Aerodontalgia
- [B]: Barosinusitis (sinus barotrauma)
- [C]: The Staggers
- [D]: Hyperventilation syndrome

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Swollen mucous membranes block the sinus ostia, causing negative pressure and intense barosinusitis pain during descent.